

Beginner's Quiz: Neutron Stars

Cosmic Compass

November 16, 2025

Instructions

Answer the following multiple-choice questions about neutron stars. Each question has 4 options (A-D). Circle the correct answer. Good luck!

1. What is the primary force that supports a neutron star against gravitational collapse?

A. Electron degeneracy pressure C. Thermal pressure
B. Neutron degeneracy pressure D. Magnetic pressure

Answer: B

2. What is the typical mass range of a neutron star (in solar masses)?

A. 0.1–1.0 C. 5.0–10.0
B. 1.4–3.0 D. 20.0–50.0

Answer: B

3. How large is the radius of a typical neutron star?

A. 1–5 km C. 100–150 km
B. 10–15 km D. 1000–1500 km

Answer: B

4. What is the density of a neutron star compared to an atomic nucleus?

A. Less dense C. Much denser
B. Similar D. Impossible to compare

Answer: C

5. What observational phenomenon led to the discovery of neutron stars?

- A. Gamma-ray bursts
- B. Pulsating radio waves (pulsars)
- C. Supernova remnants
- D. X-ray binaries

Answer: B

6. What is the surface temperature of a newly formed neutron star (approximately)?

- A. 1,000 K
- B. 1 million K
- C. 1 billion K
- D. 100 billion K

Answer: D

7. What happens if a neutron star's mass exceeds 3 solar masses?

- A. It explodes
- B. It becomes a white dwarf
- C. It collapses into a black hole
- D. It stops rotating

Answer: C

8. What is the outer crust of a neutron star primarily made of?

- A. Hydrogen
- B. Iron and nickel
- C. Silicon
- D. Carbon

Answer: B

9. Why do neutron stars spin rapidly?

- A. They are born spinning
- B. Conservation of angular momentum
- C. Magnetic fields accelerate them
- D. External forces push them

Answer: B

10. What is a "kilonova"?

- A. A supernova from a massive star
- B. A merger of two neutron stars
- C. A black hole formation
- D. A white dwarf explosion

Answer: B

Visual Aids

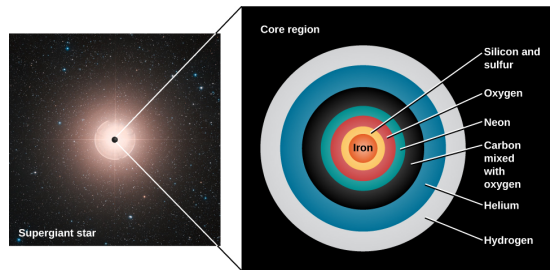


Figure 1: Artist's depiction of a neutron star.

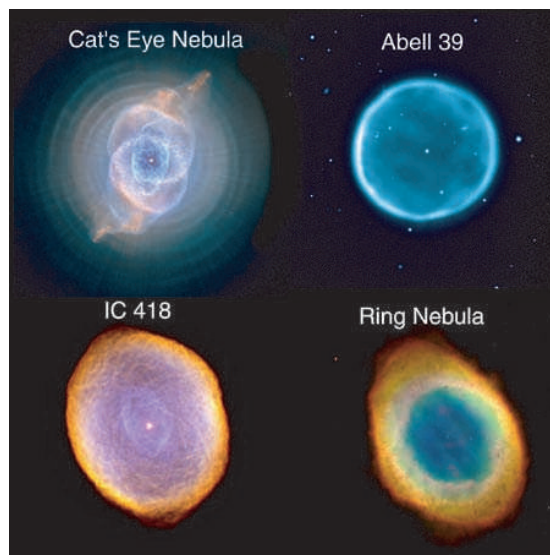


Figure 2: Cross-section of a neutron star's layers.